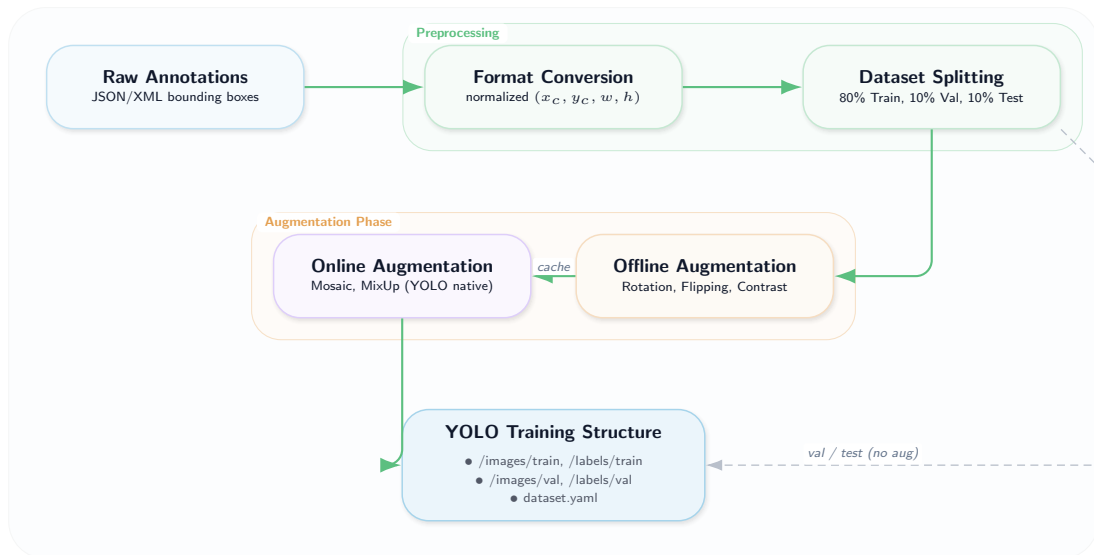


YOLO Data Pipeline Flow

Conversion of annotations to YOLO format, train/val/test splitting, and online/offline image augmentations for robust object detection.



Augmentation Strategy — Offline augmentation expands the static dataset to combat class imbalance. Online augmentation (Mosaic, MixUp) is applied dynamically during YOLO training by the dataloader to improve model generalization and detection robustness on variable scales.